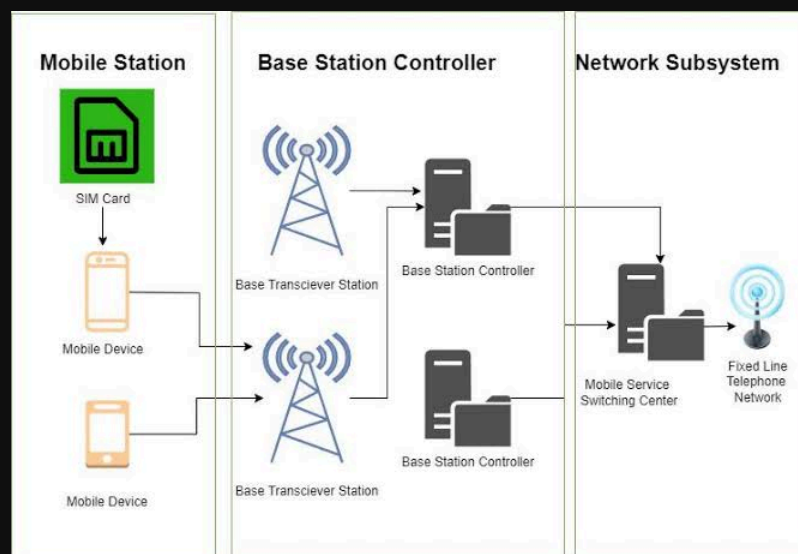


Here is a structured, exam-oriented response designed for maximum scoring in a university theoretical paper.

1. GSM Subsystems & Base Station Subsystem (BSS)

The Global System for Mobile Communications (GSM) architecture is divided into three primary functional subsystems, complemented by the user end:

- **Mobile Station (MS):** Comprises the physical mobile device (ME) and the Subscriber Identity Module (SIM) card.
- **Base Station Subsystem (BSS):** Handles radio transmission, radio channel management, and radio signal control.
- **Network and Switching Subsystem (NSS):** Manages call processing, routing, subscriber data, and mobility functions.
- **Operation and Support Subsystem (OSS):** Responsible for network management, performance monitoring, billing support, and maintainability.



GSM Subsystems Overview. Source: GeeksforGeeks

Components of the Base Station Subsystem (BSS)

The BSS links the Mobile Station to the Network Switching Subsystem (NSS) across the **Um (Air) Interface**. It contains two main physical components:

A. Base Transceiver Station (BTS)

The BTS houses the radio equipment needed to communicate across a single cell or sector.

- **Radio Transceivers (TRX):** Transmits and receives radio signals via the Um interface.
- **Signal Processing:** Handles channel encoding, encryption, modulation (GMSK), and signal quality measurements.
- **Antenna System:** Configured for omnidirectional or directional coverage.

B. Base Station Controller (BSC)

The BSC acts as the central brain for multiple BTSs, managing radio resource allocation.

- **Radio Channel Management:** Allocates and frees frequencies and timeslots.
- **Handover Execution:** Manages radio-frequency handovers between different cells/BTSs within its control.
- **Power Control:** Dynamically adjusts transmitter power levels for both the BTS and MS to minimize interference.
- **Transcoding & Rate Adaptation Unit (TRAU):** Converts 64 kbps PCM voice signals from the NSS down to the 13 kbps full-rate voice coding used over the radio link.

2. HLR vs. VLR: Subscriber Data & Mobility Management

The Home Location Register (HLR) and Visitor Location Register (VLR) are central data repositories within the NSS. Their functional differences are outlined below:

Feature / Metric	Home Location Register (HLR)	Visitor Location Register (VLR)
Nature & Scope	Centralized, permanent database per mobile network operator.	Local, temporary database integrated directly with an MSC.
Database Type	Permanent (Master store for network subscribers).	Dynamic / Temporary (Holds data only while MS is in its area).

Feature / Metric	Home Location Register (HLR)	Visitor Location Register (VLR)
Data Stored	IMSI, MSISDN, subscribed services, authentication keys, and current VLR address .	Temporary Mobile Subscriber Identity (TMSI), IMSI, MSISDN, and current Location Area Code (LAC) .
Mobility Role	Tracks which VLR region the subscriber is currently visiting worldwide.	Tracks precise cell location (LAC) for active paging and local switching.
Record Creation	Configured during service activation or SIM issuance.	Created dynamically when a mobile device enters the MSC/VLR area via a Location Update procedure.
...		

Roles in Mobility Management

1. **Location Registration:** When an MS enters a new location area, the local **VLR** issues a **Location Update** request to the subscriber's **HLR**. The HLR updates its pointer to the new VLR and instructs the old VLR to delete the stale temporary profile.
2. **Paging:** When a call arrives, the **HLR** identifies the correct **VLR**. The VLR then uses its local records (Location Area Code) to issue a paging request across the specific cell sites under its jurisdiction.

3. Roles of MSC and GMSC in Call Routing

Mobile Switching Center (MSC)

The MSC is the core switching node of the GSM Network Subsystem (NSS).

- **Call Control & Switching:** Handles call setup, routing, signaling, and call tear-down for subscribers in its serving area.
- **Inter-BSC Handovers:** Coordinates handovers when a mobile user moves between two BSCs attached to the same MSC.
- **Billing & Accounting:** Collects Call Detail Records (CDRs) for billing.
- **Local VLR Interface:** Consults the integrated VLR to retrieve temporary subscriber parameters (like TMSI and current Location Area) for incoming and outgoing calls.

Gateway MSC (GMSC)

The GMSC serves as the specialized edge router and interface point between the GSM network and external fixed/other mobile networks (such as PSTN, ISDN, or PLMN).

- **External Interfacing:** Receives incoming calls originated from fixed networks (like landlines).
- **Interrogation of HLR:** Extracted from the dialed telephone number (MSISDN), the GMSC performs a database lookup query to the subscriber's HLR to fetch the **Mobile Station Roaming Number (MSRN)**.
- **Routing Execution:** Uses the retrieved MSRN to forward the call directly to the target MSC/VLR currently serving the destination mobile device.

Step-by-Step Call Routing Scenario (Landline PSTN to GSM Mobile)

1. **Initiation:** A landline caller dials a GSM phone number (**MSISDN**).
2. **External Transit:** The PSTN routes the call directly to the GSM operator's **Gateway MSC (GMSC)**.
3. **HLR Interrogation:** The GMSC queries the subscriber's **HLR** using the MSISDN.
4. **MSRN Retrieval:** The HLR contacts the destination **VLR** to obtain a temporary **Mobile Station Roaming Number (MSRN)** and returns it to the GMSC.
5. **Routing to Target MSC:** The GMSC uses the MSRN to route the call through the network to the **Serving MSC**.
6. **Paging & Setup:** The Serving MSC asks its local **VLR** for the subscriber's Location Area Code (LAC), commands the **BSC/BTS** to page the Mobile Station, and completes the connection upon answer.

Would you like to explore related GSM examination topics?

Explain GSM Call Setup & Handover Types



Compare GSM, 3G (UMTS), and 4G (LTE) Architectures

